

Week 1

Implement vacuum cleaner agent

```
import random
import time

class VacuumEnvironment:
    def __init__(self):

        self.locations = {
            "A": random.choice(["Clean", "Dirty"]),
            "B": random.choice(["Clean", "Dirty"])
        }

        self.vacuum_location = random.choice(["A", "B"])

    def display_status(self):
        print(f"Status -> Room A: {self.locations['A']} | Room B: {self.locations['B']}")
        print(f"Pos-> Room {self.vacuum_location}\n")

class ReflexVacuumAgent:
    def __init__(self):
        pass

    def program(self, location, status):

        if status == "Dirty":
            return "Suck"
        elif location == "A":
            return "Right"
        elif location == "B":
            return "Left"

def run_simulation(steps=4):

    env = VacuumEnvironment()
```

```

agent = ReflexVacuumAgent()

env.display_status()

for step in range(1, steps + 1):
    print(f"--- Step {step} ---")
    current_loc = env.vacuum_location
    current_status = env.locations[current_loc]

    action = agent.program(current_loc, current_status)
    print(f"P: (Location: {current_loc}, Status: {current_status}) ->
Action: {action}")

    if action == "Suck":
        env.locations[current_loc] = "Clean"
        print(f"Success: Cleaned Room {current_loc}.")
    elif action == "Right":
        env.vacuum_location = "B"
        print("Success: Moved Right to Room B.")
    elif action == "Left":
        env.vacuum_location = "A"
        print("Success: Moved Left to Room A.")

    env.display_status()
    time.sleep(1)

    if env.locations["A"] == "Clean" and env.locations["B"] ==
"Clean":
        print("Goal Achieved: Both rooms are perfectly clean")
        break

if __name__ == "__main__":
    run_simulation()

```

Output

```
if __name__ == "__main__":
    run_simulation()

• Status -> Room A: Dirty | Room B: Dirty
  Pos-> Room B

  --- Step 1 ---
  P: (Location: B, Status: Dirty) -> Action: Suck
  Success: Cleaned Room B.
  Status -> Room A: Dirty | Room B: Clean
  Pos-> Room B

  --- Step 2 ---
  P: (Location: B, Status: Clean) -> Action: Left
  Success: Moved Left to Room A.
  Status -> Room A: Dirty | Room B: Clean
  Pos-> Room A

  --- Step 3 ---
  P: (Location: A, Status: Dirty) -> Action: Suck
  Success: Cleaned Room A.
  Status -> Room A: Clean | Room B: Clean
  Pos-> Room A

  Goal Achieved: Both rooms are perfectly clean
```